

FRONTEND SPECIFICATION DOCUMENT

Corporate Finance Management System

Dashboard · Admin Panel · AI Assistant · Analytics · Power BI & Presentation Studio · Share Tracker · Employee Directory

Version 1.0 | Prepared for single-company internal deployment | July 2026

1. Product Overview

This document specifies the frontend architecture, design system, page-level UX, and third-party integration contracts for a full Finance Management System built for a single company's internal finance operations. The system gives finance staff and admins a single pane of glass to monitor turnover and profit/loss, chat with an AI assistant for financial queries, generate Power BI reports and presentations from templates, track the company's own share price against competitor benchmarks, and manage the finance department's employee roster.

1.1 Core Modules

Module	Purpose
Dashboard	KPI overview, bar/pie charts for turnover & P&L (daily/monthly/yearly), quick AI suggestions widget
Admin Panel	User/role management, module access control, audit log, system settings
AI Chat Assistant	Conversational finance assistant — Q&A on company data, forecasting, anomaly explanations
Suggestions & Performance	AI-generated recommendations, KPI performance scoring, trend flags
Reports Studio (Power BI + Presentations)	Generate Power BI reports and slide decks from a template gallery
Share Value Tracker	Live/near-live share price of the company + peer comparison board
Employee Directory (Finance Dept.)	List employees, details, designation, department hierarchy

1.2 Primary User Roles

Role	Access
Super Admin	Full access incl. Admin Panel, user provisioning, integration keys
Finance Manager	Dashboard, Reports Studio, Share Tracker, Employee Directory (read/write within finance)
Finance Analyst	Dashboard (view), AI Chat, Reports Studio (generate only)
Viewer/Executive	Dashboard (view-only), Share Tracker, Suggestions summary

2. Design System

2.1 Color Palette

Token	Hex	Usage
 Primary / Navy	#0B1F3A	Sidebar, headers, primary text on light bg
 Primary Accent / Blue	#2F6FED	Primary buttons, links, active states, chart series 1
 Success / Profit Green	#1FBF75	Positive turnover, profit indicators, upward trend
 Danger / Loss Red	#E5484D	Loss indicators, downward trend, destructive actions
 Warning / Amber	#F5A623	Alerts, pending states, AI-flagged anomalies
 Secondary / Violet	#7C5CFC	AI Chat accents, chart series 2
 Surface / Background	#F7F9FC	App background
 Card Surface	#FFFFFF	Cards, modals, table rows
 Border / Divider	#D7DCE3	Borders, dividers, disabled fields
 Text Secondary	#4A5568	Subtext, labels, placeholders
 Text Primary	#1A1F2B	Body text

Chart series palette (multi-series bar/pie): #2F6FED, #1FBF75, #F5A623, #7C5CFC, #E5484D, #14B8C4 — cycle in this order for consistency across all chart components.

2.2 Typography

Role	Font	Size / Weight
Font family	Inter (fallback: 'Segoe UI', Roboto, sans-serif)	—
H1 / Page title	Inter	28px / 700
H2 / Section title	Inter	20px / 600
H3 / Card title	Inter	16px / 600
Body	Inter	14px / 400
Small / Caption	Inter	12px / 400, color #4A5568

KPI Numbers	Inter / tabular-nums	32px / 700
Monospace (codes, IDs)	JetBrains Mono	13px / 400

2.3 Spacing & Layout Grid

- Base spacing unit: **4px**. Standard scale: 4, 8, 12, 16, 24, 32, 48, 64px.
- App shell: fixed left sidebar 260px (collapsible to 72px icon rail) + top bar 64px + fluid content area, max content width 1440px, 24px gutter.
- Grid: 12-column responsive grid, 24px gutters. Breakpoints — mobile <640px, tablet 640-1024px, desktop 1024-1440px, wide >1440px.
- Card padding: 20px (desktop), 16px (tablet/mobile). Card border-radius: 12px.
- Section vertical rhythm: 32px between major blocks, 16px between related cards.

2.4 Component Styles

Buttons

Variant	Style
Primary	bg #2F6FED, text #FFFFFF, radius 8px, height 40px, padding 0 20px, hover bg #1E5BD6, disabled bg #B8CCF7
Secondary	bg transparent, border 1px #2F6FED, text #2F6FED, hover bg #EEF3FF
Destructive	bg #E5484D, text #FFFFFF, hover bg #C93E42
Ghost/Text	bg transparent, text #2F6FED, no border, hover underline

Inputs

Height 40px, radius 8px, border 1px #D7DCE3, padding 0 12px, font 14px. Focus: border #2F6FED + 3px outer glow rgba(47,111,237,0.15). Error: border #E5484D + helper text 12px #E5484D below field. Label: 12px 600 #4A5568, 4px above input.

Cards

bg #FFFFFF, radius 12px, border 1px #E7EAF0, shadow 0 1px 3px rgba(16,24,40,0.06). KPI cards include: icon (24px, tinted circle bg), label (12px #4A5568), value (32px 700), delta badge (green/red pill, e.g. "+4.2%").

Modals

Max width 560px (standard) / 880px (wide, e.g. report template picker). Radius 16px, overlay rgba(11,31,58,0.5), header 20px 600 with close icon top-right, footer right-aligned action buttons, body max-height 70vh scrollable.

Tables

Row height 48px, header bg #F5F7FB text #4A5568 12px 600 uppercase letter-spacing 0.4px, zebra striping optional, sticky header on scroll, row hover bg #F7F9FC.

Charts (Bar & Pie)

Bar chart: rounded top corners 4px, bar gap 40%, gridlines #EEF1F5 dashed, tooltip dark navy bg #0B1F3A text white radius 6px. Pie/Donut: donut inner radius 60%, segment gap 2px, center label shows total, legend right-aligned with color dot + label + %.

3. Information Architecture & Page Specs

3.1 Sidebar Navigation

Dashboard · Analytics & Suggestions · AI Chat · Share Tracker · Reports Studio (Power BI / Presentations) · Employee Directory · Admin Panel · Settings

3.2 Dashboard Page

- **KPI row** (4 cards): Total Turnover (period), Net Profit/Loss, Growth % vs previous period, Active Employees count.
- **Turnover / P&L Chart block**: toggle tabs — Daily / Monthly / Yearly. Chart-type switch — Bar / Pie. Bar shows turnover vs profit/loss side-by-side per period bucket; Pie shows profit vs loss vs cost proportion for the selected period.
- **AI Suggestions widget** (compact card, 3 bullet insights, "View all" → Suggestions page).
- **Recent activity feed**: last 10 finance events (invoice posted, report generated, share alert).

3.3 Admin Panel

- User management table (name, email, role, status, last login) with invite/edit/deactivate actions.
- Role & permission matrix editor (checkbox grid: module × permission).
- Integration settings: API keys for Power BI, market-data provider, AI provider (masked input + "Test connection" button).
- Audit log table: timestamp, user, action, module.

3.4 AI Chat Page

- Left: conversation history list. Center: chat thread (user bubble right, right-aligned #2F6FED bg white text; AI bubble left, #F5F7FB bg #1A1F2B text, avatar violet #7C5CFC circle).
- Input bar: text field + attach (for uploading a statement) + send button; suggested prompt chips above input (e.g. "Summarize this month's P&L", "Why did Q2 turnover dip?").
- AI responses can render inline mini-charts when the answer references turnover/P&L data.

3.5 Suggestions & Performance Page

- Performance scorecards per KPI (turnover growth, expense ratio, margin trend) with a 0–100 score gauge.
- AI recommendation list: card per suggestion — title, rationale, impact estimate, priority badge (High/Med/Low), "Mark actioned" toggle.
- Trend flags: anomaly banner (amber) when a metric deviates >2 standard deviations from trailing average.

3.6 Reports Studio — Power BI & Presentations

- **Template gallery** (grid of cards, thumbnail + name + tag: "Power BI" or "Presentation"): Monthly P&L Report, Board Deck, Quarterly Turnover Review, Investor Summary, Department Cost Breakdown, etc.
- Selecting a template opens a config modal (wide, 880px): pick date range, metrics to include, branding options → "Generate".
- Generated Power BI reports embed in an iframe panel with export (PDF/PPTX) and "Open in Power BI" link.
- Generated presentations show a slide thumbnail strip + preview pane + Download (.pptx/.pdf) button.

3.7 Share Value Tracker

- Header: company name, ticker, current price, day change (colored delta), last updated timestamp.
- Price chart: line/candlestick, range selector (1D/1W/1M/1Y/5Y).
- **Comparison board**: table + overlay line chart comparing company share performance (% change) against 2–5 selected peer companies over the same range; each peer row shows ticker, price, %change, add/remove control.

3.8 Employee Directory (Finance Department)

- Searchable/filterable table: Name, Photo, Employee ID, Designation, Sub-team, Email, Phone, Date Joined, Status.
- Row click → side drawer with full profile: reporting manager, tenure, past designations, contact.
- Filter by designation (CFO, Finance Manager, Senior Analyst, Analyst, Accounts Executive, Intern).

4. API & Third-Party Integration Specification

Below is the full integration contract for every external service the frontend depends on. All requests are proxied through the app's backend (Node/Express or similar) — the frontend never calls third-party APIs directly with raw secret keys; it calls the internal API which forwards to these services.

4.1 AI Chat — Anthropic Claude API

Field	Detail
Purpose	Powers the AI Chat module and AI-generated Suggestions using company financial data as context
Endpoint	<code>POST https://api.anthropic.com/v1/messages</code>
Auth	Header <code>x-api-key</code> (server-side only, never exposed to frontend)
Request body	model, max_tokens, messages[] (role/content), optional system prompt containing scoped financial context
Response	content[] blocks (type: text), usage tokens — rendered as chat bubble in UI
Frontend contract	Internal endpoint <code>POST /api/ai/chat</code> → { conversationId, message } → returns { reply, suggestedCharts? }

4.2 Market Data — Share Price & Peer Comparison

Field	Detail
Purpose	Fetches live/delayed share price for the company and peer companies
Provider options	Alpha Vantage, NSE/BSE data API (India), or Yahoo Finance-compatible feed — pick one per deployment region
Sample endpoint (Alpha Vantage)	<code>GET https://www.alphavantage.co/query?function=GLOBAL_QUOTE&symbol=TICKER&apikey=KEY</code>
Data sent	Ticker symbol, function type (quote/daily/intraday)
Response expected	{ price, change, changePercent, volume, latestTradingDay }
Frontend contract	<code>GET /api/shares/:ticker</code> and <code>GET /api/shares/compare?tickers=A,B,C</code>
Polling	Client polls every 60s during market hours (configurable); falls back to "last updated" timestamp when market closed

4.3 Power BI Embedding & Report Generation

Field	Detail
Purpose	Generate and embed Power BI reports from templates using live dashboard data
Service	Power BI REST API + Power BI Embedded (Azure AD app registration required)
Auth	OAuth2 client-credentials flow against Azure AD → bearer token
Key endpoints	<code>POST https://login.microsoftonline.com/{tenant}/oauth2/v2.0/token</code> (get token) <code>GET https://api.powerbi.com/v1.0/myorg/groups/{groupId}/reports</code> (list reports) <code>POST .../reports/{reportId}/GenerateToken</code> (embed token) <code>POST .../datasets/{datasetId}/refreshes</code> (push latest turnover/P&L data before generating)
Data sent	Dataset refresh payload (turnover, P&L rows), report template ID, embed config
Response expected	embedUrl, embedToken, expiry — used to render <code><iframe></code> / Power BI JS SDK embed
Frontend contract	<code>POST /api/reports/powerbi/generate</code> { templateId, dateRange, metrics } → { embedUrl, embedToken, reportId }

4.4 Presentation Generation

Field	Detail
Purpose	Generate downloadable slide decks (board decks, investor summaries) from templates
Approach	Server-side generation via PptxGenJS (Node) or Google Slides API (if cloud-based editing desired)
Google Slides API sample endpoint	<code>POST https://slides.googleapis.com/v1/presentations</code> (create) and <code>POST .../presentations/{id}:batchUpdate</code> (populate slides)
Data sent	Template ID, slide content blocks (KPI values, chart images as base64/PNG, narrative text from AI suggestions)
Response expected	presentationId + exportable link, or direct .pptx binary stream for download
Frontend contract	<code>POST /api/reports/presentation/generate</code> { templateId, sections[] } → { downloadUrl, previewSlides[] }

4.5 Authentication & User Management

Field	Detail
Purpose	Login, session, role-based access for Admin Panel
Provider	Auth0 or Firebase Authentication (email/password + optional SSO)
Key endpoints	<code>POST /oauth/token</code> (Auth0 login), <code>GET /userinfo</code>

Data sent	email, password (or SSO assertion)
Response expected	access_token (JWT), id_token, expiry, user profile claims (role, department)
Frontend contract	JWT stored in httpOnly cookie via backend session; frontend reads role from <code>/api/auth/me</code>

4.6 File Storage — Charts, Exports, Employee Photos

Field	Detail
Purpose	Store generated PPTX/PDF exports and employee photos
Provider	AWS S3 (or Firebase Storage)
Key endpoints	<code>PUT https://{bucket}.s3.{region}.amazonaws.com/{key}</code> (pre-signed URL upload), <code>GET</code> (pre-signed download)
Data sent	File binary + pre-signed URL request from backend
Response expected	200 OK + object URL; frontend receives signed download link with expiry
Frontend contract	<code>GET /api/files/:id/signed-url</code>

4.7 Notifications (Optional — Alerts on Share Price / Anomalies)

Field	Detail
Purpose	Email/SMS alert when share price crosses threshold or AI flags a P&L anomaly
Provider	SendGrid (email) / Twilio (SMS)
Key endpoint	<code>POST https://api.sendgrid.com/v3/mail/send</code>
Data sent	to, from, subject, dynamic template data (metric name, threshold, current value)
Response expected	202 Accepted
Frontend contract	Alert rules configured in Admin Panel → <code>POST /api/alerts/rules</code>

4.8 Charting Library (Client-Side, Not a Network API)

Recharts or Chart.js renders all bar/pie/line charts client-side from data returned by the internal `/api/finance/turnover` and `/api/finance/pnl` endpoints (query params: `period=daily|monthly|yearly`, `from`, `to`). No external network call is made for rendering — only for data fetch from the internal backend.

Security note: every third-party API key (Power BI, market data, AI provider, SendGrid) must live server-side only (env vars / secrets manager), accessed via the internal API layer. The frontend never holds a raw third-party key.

5. Non-Functional Notes

- Accessibility: WCAG 2.1 AA — min contrast 4.5:1 for text, all charts have data-table fallback for screen readers.
- Performance: dashboard first meaningful paint < 2.5s on 4G; chart data paginated/aggregated server-side for yearly views.
- Responsiveness: sidebar collapses to icon rail below 1024px; tables convert to stacked cards below 640px.
- Theming: dark mode variant using inverted surface tokens (`#0B1F3A` bg, `#F7F9FC` text) — same accent colors retained.

End of document. This specification covers frontend architecture, design tokens, page-level UX, and integration contracts. Backend schema, database design, and DevOps/infra are out of scope for this document.